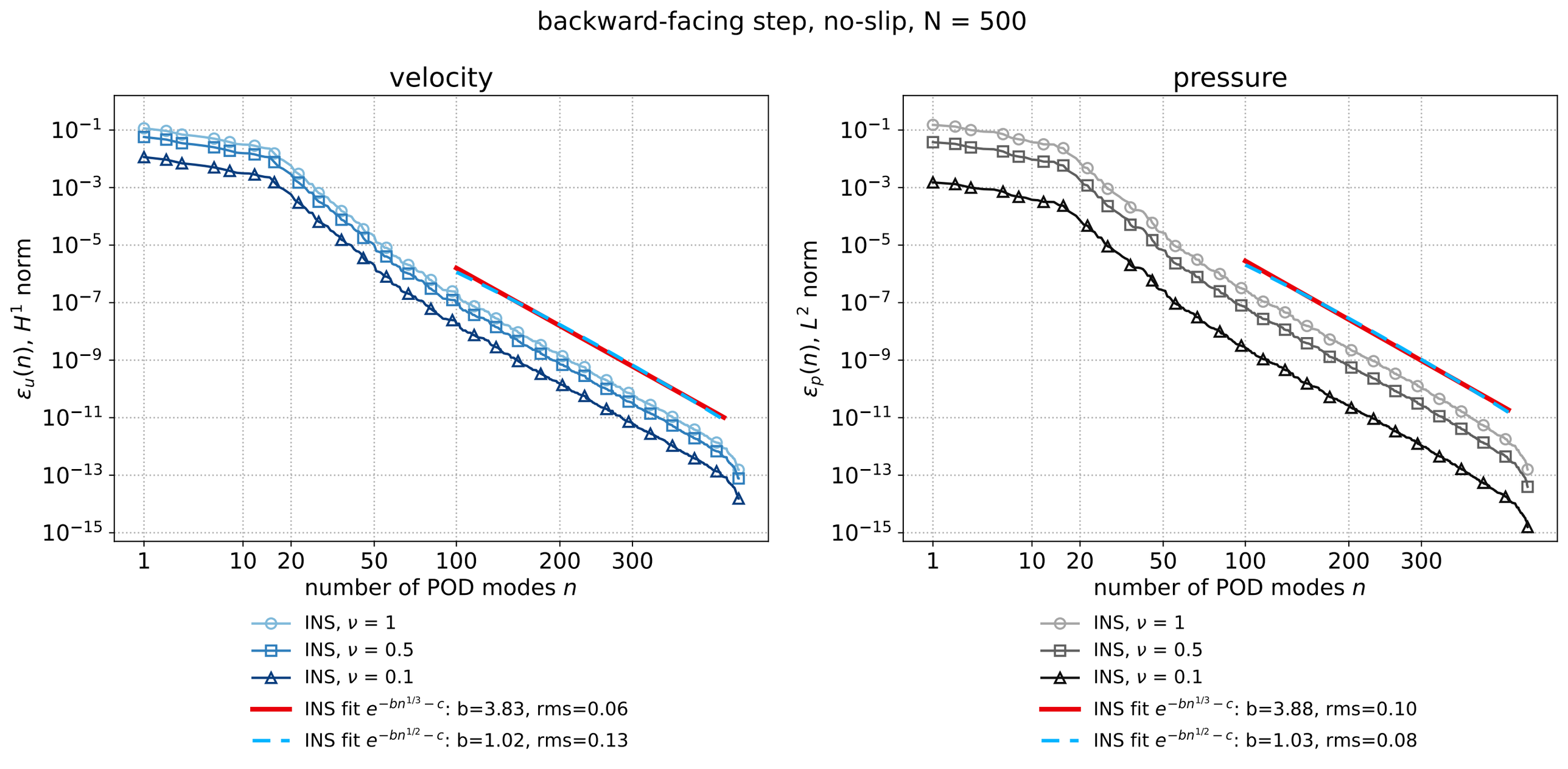
backward-facing step, no-slip, N = 500
velocity
pressure
$\varepsilon_u(n)$, H^1 norm
$\varepsilon_p(n)$, L^2 norm
10^{-1}
10^{-3}
10^{-5}
10^{-7}
10^{-9}
10^{-11}
10^{-13}
10^{-15}
1
10
20
50
100
200
300
number of POD modes n
INS, ν = 1
INS, ν = 0.5
INS, ν = 0.1
INS fit $e^{-bn^{1/3}-c}$: b=3.83, rms=0.06
INS fit $e^{-bn^{1/2}-c}$: b=1.02, rms=0.13
INS fit $e^{-bn^{1/3}-c}$: b=3.88, rms=0.10
INS fit $e^{-bn^{1/2}-c}$: b=1.03, rms=0.08